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- **Record different configurations of the core utilization, clock period, DRC violations, slack, chip area, and wire length. Furthermore, explain the observation of the experiment. (Try more than 5 different configurations to get full points in this question.)**

Core Utilization	Clock Period	DRC violations	Slack	Chip Area	Wire length
0.5	700	0	19.2	50916.999	188107.1440
0.6	700	0	22.523	42877.797	186078.6040
0.4	600	0	5.169	62888.348	187162.8200
0.4	500	6	-48.7	62888.348	198223.9280
0.45	700	0	26.029	56247.914	184833.3960
0.65	700	0	31.084	39775.298	179842.2840
0.65	700	1	7.007	37121.981	175404.2880
0.65	556	0	3.7	39775.298	180461.0960
0.65	553	0	2.1	39775.298	180690.1120
0.67	595	0	5.648	38664.014	178221.3040

- **Explain how the adjustments of the clock period and the core utilization affect the metrics (DRC violations, slack, chip area, and wire length).**

DRC (Design Rule Check) Violations:

A tighter (shorter) clock period may force cells to be placed closer together to meet timing constraints, increasing routing congestion and potential DRC violations (e.g., spacing and routing violations).

A relaxed (longer) clock period allows for a more spread-out placement, reducing congestion and potential violations.

Slack:

Reducing the clock period (higher frequency) decreases slack, increasing the risk of timing violations.

Increasing the clock period gives more room for signal propagation, improving timing slack.

Chip Area:

A shorter clock period may require larger drive strength cells, additional buffers, and more complex logic optimizations, leading to increased area.

A longer clock period allows for simpler logic and fewer buffers, reducing

chip area.

Wire Length:

A reduced clock period often results in denser placement to minimize delays, increasing wire congestion and length.

A relaxed clock period leads to more distributed placement, which can reduce wire congestion.

High Core Utilization (>80%):

Increases DRC violations due to congestion and limited routing resources.

Reduces slack as signals face more routing detours and higher delays.

Minimizes chip area, making the design compact but harder to optimize.

Increases wire length due to congestion and suboptimal placement.

Low Core Utilization (<60%):

Reduces DRC violations by allowing better cell placement and routing.

Increases slack as signal paths are more optimized.

Increases chip area due to excessive whitespace but provides room for future optimizations.

Reduces wire length by allowing shorter and more direct interconnects.

- **Explain the purpose of inserting well tap cell**.

Well tap cells are inserted in the design to prevent latch-up, which can cause functional failures in CMOS circuits. Their key roles include:

Preventing Latch-up:

Well tap cells connect the N-well to VDD and P-well to GND, preventing unintended parasitic transistor activation.

Maintaining Proper Well Potential:

Ensures that the well regions are properly biased, reducing the risk of voltage drops that can affect transistor performance.

Enhancing Reliability:

Reduces substrate noise and ensures consistent electrical performance across the chip.

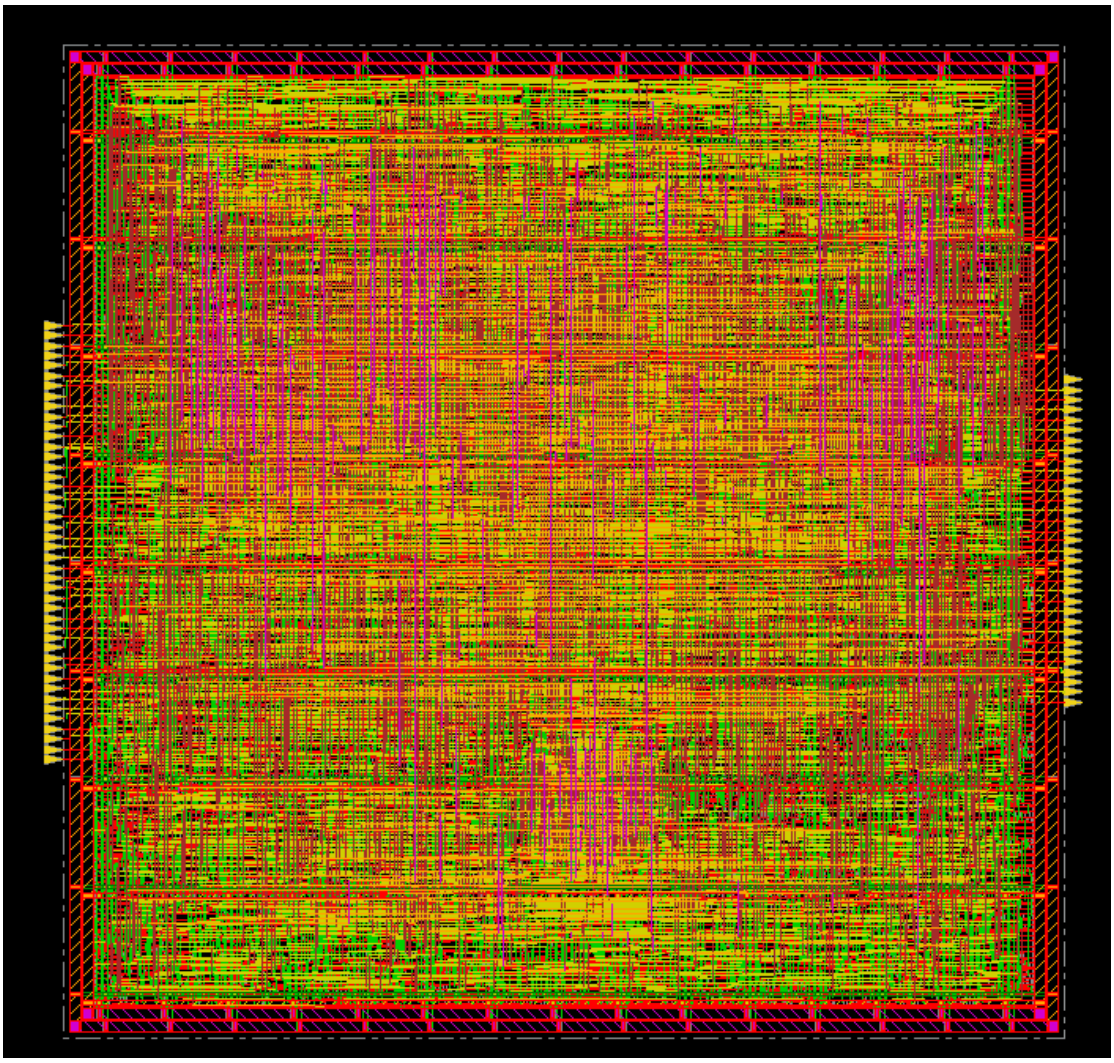
- **Show the configuration from your best result. The information to be included in the configuration is listed below. Please try to maintain nonnegative slack and zero DRC violations.**

  i. Clock period (in file sha256.sdc) 553

  ii. Slack time (in generated file timing.rpt) 2.1

  iii. Total area of chip (in generated file summary.rpt) 39775.298

  iv. Total wire length (in generated file summary.rpt) 180690.1120

  v. DRC violations (in generated file drc.rpt). 0

- **Show the final chip layout of your best result generated by Innovus (use printscreen to save the final layout and paste on the report)**